COUNTERPARTY CREDIT RISK

Risk Management and Financial Institutions

Lecture 10.

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Bilateral vs multilateral clearing

Regulation of OTC derivatives market

CVA, wrong-way risk

Default intesity, hazard rate

Black model

Counterparty credit risk

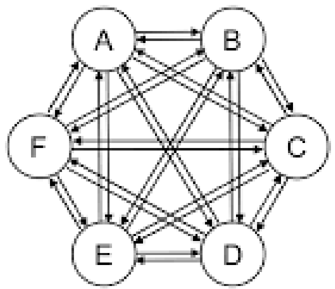
Counterparty credit risk (CCR) is the risk that the **counterparty to a transaction** could default before the final settlement of the transaction's cash flows. An economic loss would occur if the transactions or portfolio of transactions with the counterparty has a positive economic value at the time of default. Unlike a firm's exposure to credit risk through a loan, where the exposure to credit risk is unilateral and only the lending bank faces the risk of loss, **CCR creates a bilateral risk of loss**: the market value of the transaction can be positive or negative to either counterparty to the transaction. **The market value is uncertain and can vary over time** with the movement of underlying market factors. (Basel Framework)

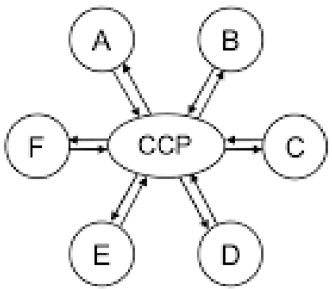
Financial Markets

- Exchange traded
  - Traditionally exchanges have used the open-outcry system, but electronic trading has now become the norm
  - Contracts are standard; there is virtually no credit risk
- Over-the-counter (OTC)
  - A network of dealers at financial institutions, corporations, and fund managers who trade directly with each other
  - Contracts can be non-standard; there is some credit risk

Exchange vs OTC market – as we used to teach.

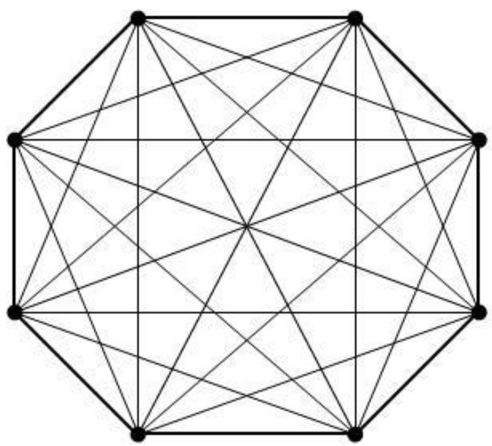
OTC market	Exchange
Bilateral clearing	Central clearing
No netting	Netting
Settlement of the profit at maturity	Daily settlement
Transactions held until maturity	Transactions are usually closed before maturity
No margin (collateral)	Margin requirements
Subject to counterparty risk	No counterparty risk



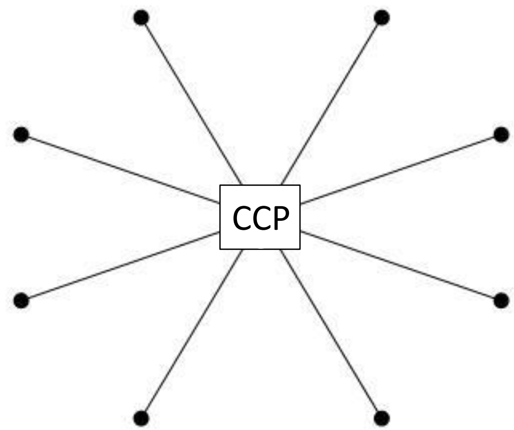


Clearing Arrangements for OTC Derivatives (Figure 6.1, page 132)

- Bilateral clearing: usually governed by an ISDA Master agreement
- Central clearing: a central counterparty (CCP) stands between the two sides



Bilateral Clearing

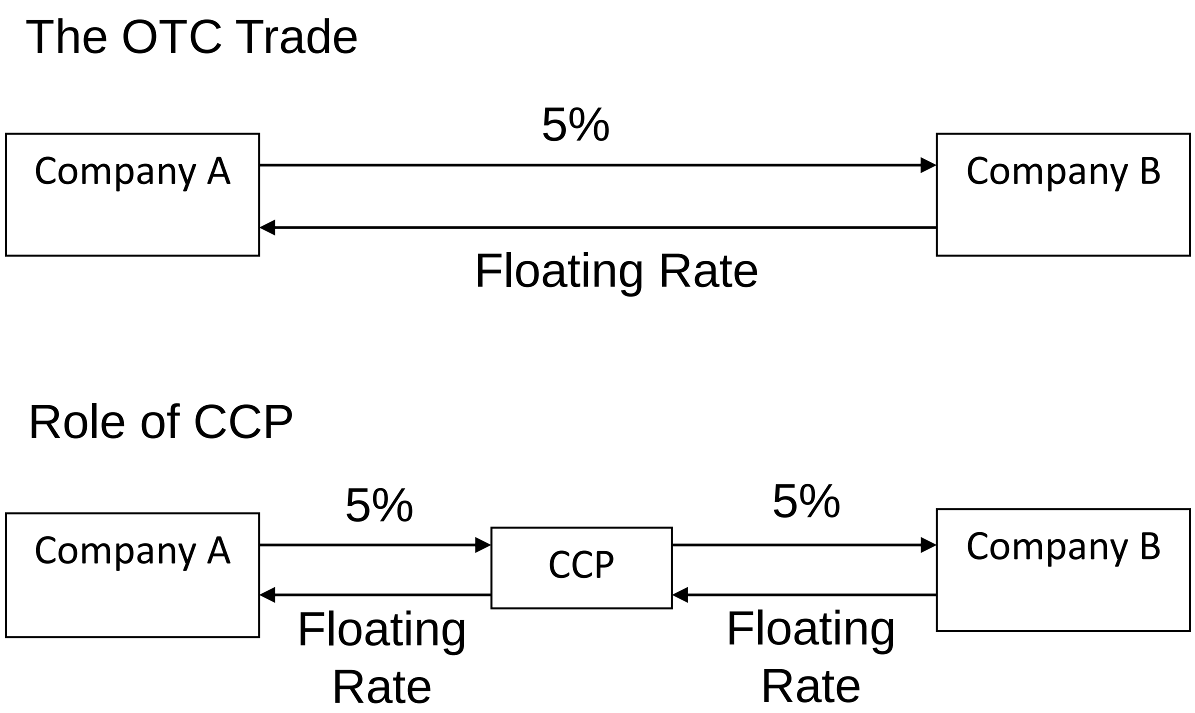


Clearing through a single CCP

Bilateral Clearing

- OTC agreement directly between two sides
- Usually governed by an ISDA master agreement
- This contains a credit support annex (CSA) identifying collateral arrangements
- Early termination provisions in the event of default not subject to usual bankruptcy rules
  - Non-defaulting party can keep collateral posted up to amount owed.
  - Claim by non-defaulting party reflects mid market values adjusted for half the bid-offer spread

Central Clearing (Figure 6.2, page 133)



Central Clearing continued

- Role of CCP is very similar to that of an exchange clearing house.
- It stands between two sides
- It requires initial margin and variation margin from its members reflecting all their transactions with CCP
- Initial margin roughly covers 5 day price moves with 99% confidence
- The members may be clearing trades for other trading entities

Variation Margin

- Margin reflecting change in value of a position
- Example: Party A and B trade. If value of outstanding positions to A increase by $X during day, then B has to provide A with $X of acceptable collateral

Initial Margin

- Even if variation margin has been posted, A may still lose money if B defaults because:
  - B is not up to date with margin payments at time of default
  - A may be subject to bid-offer spreads when it replaces transactions
- Initial margin can be used to provide additional protection

Haircuts

- Often margin is cash and interest is normally paid on cash balances
- Sometimes securities are provided instead of cash.
- There then may be a “haircut” where the market value of the securities is reduced by a certain percentage to determine their value for collateral purposes.

Netting (page 135)

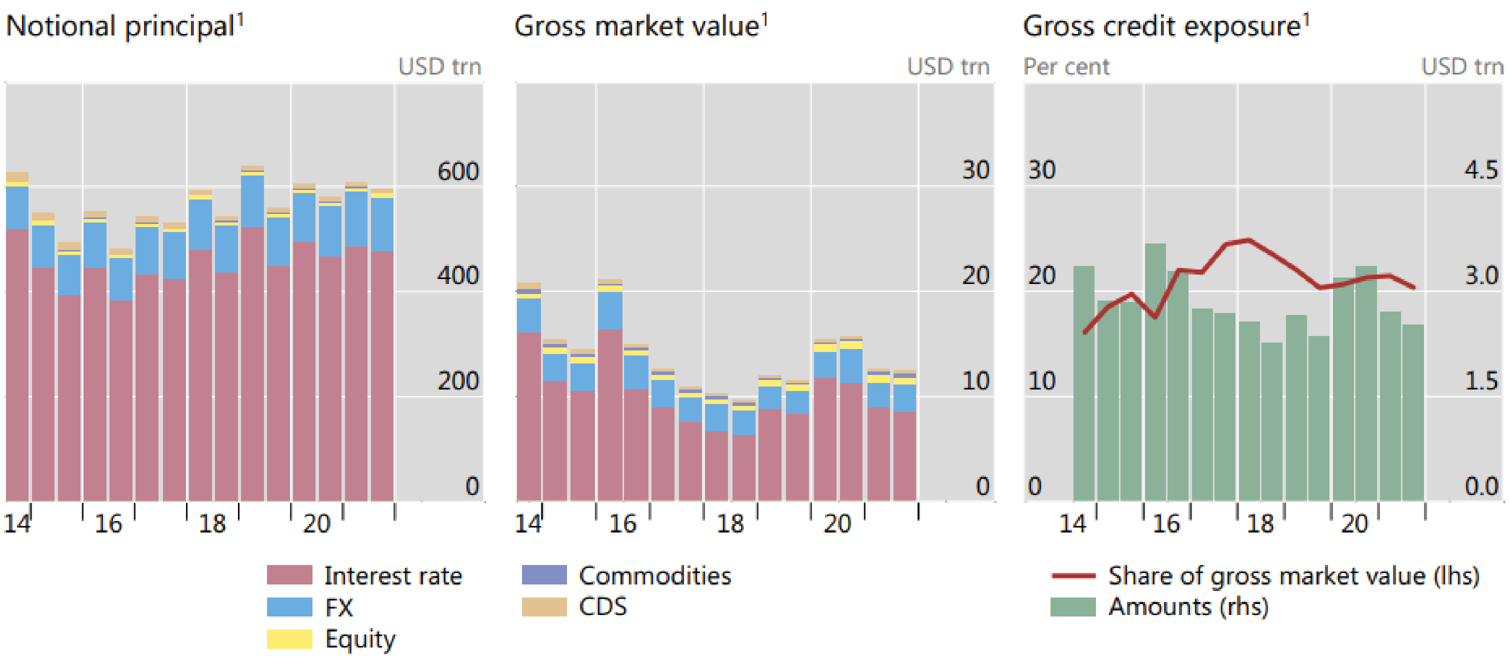
- Feature of both ISDA master agreements and CCPs
- All transactions are considered to be a single transaction…
  - in the event of default
  - for collateral calculations
- Netting reduces initial margin requirements

Waterfall for Funding Losses when Positions Have to be Closed Out (CCP or Exchange Clearing House)

- Initial margin of defaulting member
- Default fund contributions of defaulting member
- Default fund contributions of other members
- Equity of CCP/Exchange Clearing House

OTC derivative transactions

Global OTC derivatives markets — Graph A.1



[1] At half-year end (end-June and end-December). Amounts denominated in currencies other than the US dollar are converted to US dollars at the exchange rate prevailing on the reference date.

Source: BIS OTC derivatives statistics (available at www.bis.org/statistics/derstats.htm).

- OTC derivative market notional 600.000 bln USD
- Marke value 12.400 bln USD
- Exposure 2.500 bln USD

15th of September 2008

Lehman Brothers (4. largest investment bank in the US) is filed for bankruptcy protection.

(largest bankruptcy filing in the US history)

- Total asset value: 639 bln USD
- Number of derivative transactions: 930.000
- Number of ISDA contracts: 6.120
- Notional of open derivative positions: 35.000 bln USD (5% of the whole market)

Post-Crisis Regulatory Changes 1

- All standardized derivatives (i.e., plain vanilla interest rate swaps and index credit default swaps) between financial institutions must be cleared through CCPs
- Standardized derivatives between financial institutions must be traded on electronic platforms
- All OTC trades must be reported to a central trade repository

Post-Crisis Regulatory Changes 2

- Initial margin and variation margin must be posted for nonstandard trades between financial institutions (These are referred to as “uncleared” trades because they are cleared bilaterally, rather than through a CCP)
- Variation margin is transferred directly between the two sides
- Initial margin is posted with a third party

Post-Crisis Regulatory Changes 3 continued

- Initial margin for uncleared trades between financial institutions must cover 10-day price movements in stressed market conditions with 99% confidence
- Margin rules for uncleared trades were implemented starting in 2016

Calculation of Initial Margin for Uncleared Trades

- Basel Committee proposes a grid approach where initial margin is a percent of notional principal (no netting)
- ISDA proposes Standard Initial Margin Model (SIMM) which uses a version of the model building approach for handling delta and gamma risk (see Appendix M)

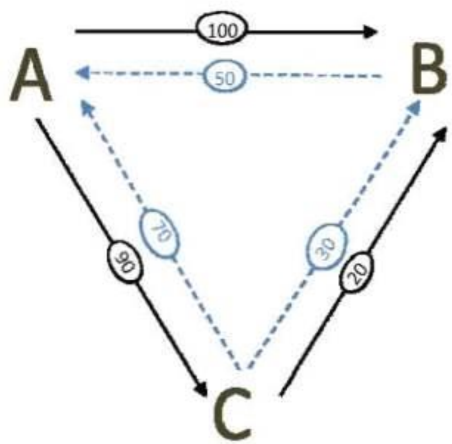
Collateral Increases

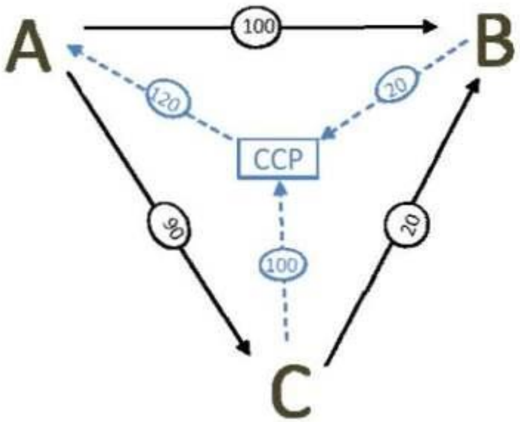
- Much more collateral will be needed to support derivatives trading
- This collateral will usually have to be cash or government securities
- This is liable to create liquidity pressures

CCPs and Netting

- CCPs should increase netting, because transactions with two different counterparties can be netted if both are cleared through the same CCP
- But it will no longer be possible to net standard with nonstandard transactions
- It is therefore conceivable that benefits of netting could decrease….

Simple Example: 3 market participants; 2 product types





Dealer	Exposure after bilateral netting
A	0
B	100
C	20
Ave	40

Dealer	Exposure after netting incl. CCP	Exposure after netting excl. CCP
A	120	0
B	120	120
C	90	90
Ave	110	70

A Convergence…

- OTC and exchange-traded derivatives markets are becoming more similar to each other as far as
  - the way trading is done
  - the way transactions are cleared
  - The collateral (margin) that has to be posted

CCPs and Too-Big-To Fail

- It can be argued that CCPs are the new TBTF financial institutions
- However, CCPs are much easier to regulate than banks and so moving risks to CCPs may make the financial system easier to regulate and therefore safer

Counterparty credit risk in the regulation

Before crises

- Contracts based on ISDA
- Netting agreement
- Credit Support Annex
  - Mutual collateral posting
  - Downgrade triggers (additional collateral, liquidation)

Basel III

- Initial and variation margin requirement for bilateral (uncleared) transactions.
- Initial margin to be posted at third party.
- Initial margin is connected to 99% 10 day VaR.
- CVA (Credit Value Adjustment)
- Wrong way risk

CVA

- Credit value adjustment (CVA) is the amount by which a dealer must reduce the value of transactions because of counterparty default risk

The CVA Calculation

Time	0	t_1	t_2	t_3	t_4	. . . .	t_n=T
Default probability		q_1	q_2	q_3	q_4	. . . .	q_n
PV of expected net exposure		v_1	v_2	v_3	v_4	. . . .	v_n

$$\text{CVA} = (1-R)\sum_{i=1}^{n} q_i v_i$$

where *R* is the recovery rate

Properties of Hazard Rates

- Suppose that $\lambda(t)$ is the hazard rate at time t
- The probability of default between times t and $t+\Delta t$ conditional on no earlier default is $\lambda(t)\Delta t$
- The probability of default by time t is

$$1-e^{-\bar{\lambda}(t)t}$$

where $\bar{\lambda}(t)$ is the average hazard rate between time zero and time t

The Default Probabilities (equation 18.3)

- The default probabilities (i.e., the q_i's) are calculated from credit spreads
- If s_i is the credit spread for maturity t_i, $\bar{\lambda}_i$ is the average hazard rate up to time t_i , and R is the recovery rate

$$\bar{\lambda}_i = \frac{s_i}{1-R} \qquad\qquad q_i = e^{-\bar{\lambda}_{i-1}t_{i-1}} - e^{-\bar{\lambda}_i t_i}$$

$$q_i = \exp\left(-\frac{s_{i-1}t_{i-1}}{1-R}\right) - \exp\left(-\frac{s_i t_i}{1-R}\right)$$

The Expected Exposures

- The v_i are calculated using Monte Carlo simulation. Random paths are chosen for all the market variables underlying the derivatives and the net exposure is calculated at the mid point of each time interval.
- v_i is the present value of the average net exposure at the ith default time

The Expected Exposure

- If no collateral is required, the exposure at each time for each random trial is max(V,0) where V is the value of the derivatives portfolio to the dealer
- If the collateral equal to C is expected to be posted by the counterparty at the time of the default (with a negative C indicating collateral expected to be posted by the dealer), the exposure is max($V - C$, 0)
- This formula reflects the fact that the dealer will have an exposure if it has posted more collateral than the value of the derivatives to the counterparty.

CVA Risk

- The CVA for a counterparty can be regarded as a complex derivative
- Increasingly dealers are managing it like any other derivative
- Two sources of risk:
  - Changes in counterparty spreads
  - Changes in market variables underlying the portfolio

Wrong Way/Right Way Risk

- Simplest assumption is that probability of default q_i is independent of net exposure v_i.
- Wrong-way risk occurs when q_i is positively dependent on v_i
- Right-way risk occurs when q_i is negatively dependent on v_i

Examples

- Wrong-way risk typically occurs when
  - Counterparty is selling credit protection
  - Counterparty is a hedge fund taking a big speculative positions
- Right-way risk typically occurs when
  - Counterparty is buying credit protection
  - Counterparty is partially hedging a major exposure

Problems in Estimating Wrong Way/Right Way Risk

- Knowing trades counterparty is doing with other dealers
- Knowing how different market variables influence the fortunes of the counterparty

Allowing for Wrong-Way risk

- One common approach is to use the “alpha” multiplier to increase the v’s
- Estimates of 1.07 to 1.1 for alpha obtained from banks
- Regulators set alpha equal to 1.4 or allows banks to use their own models, with a floor of 1.2

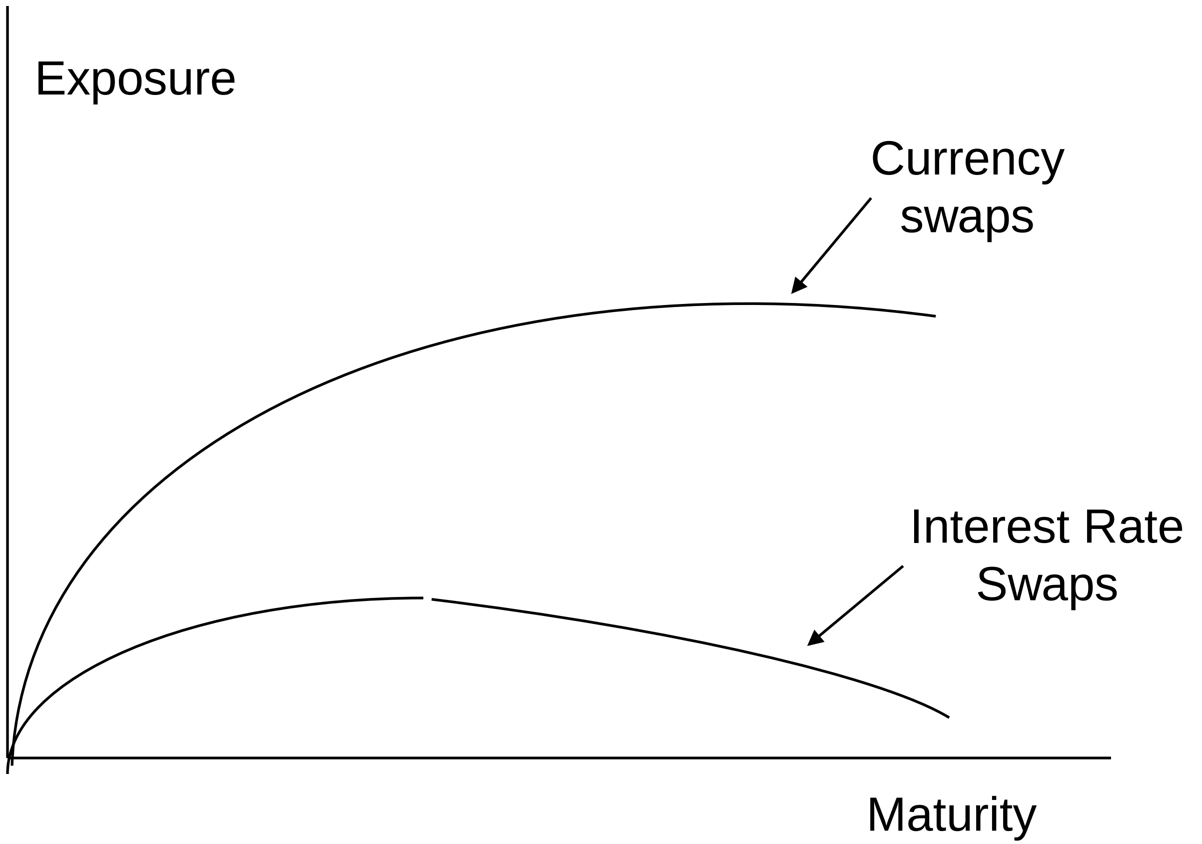
DVA

- Debit (or debt) value adjustment (DVA) is an estimate of the cost to the counterparty of a default by the dealer
- Same formulas apply except that v is counterparty's exposure to dealer and q is dealer's probability of default
- Accounting value of transactions with counterparty = No default value – CVA + DVA

DVA continued

- What happens to the reported value of transactions as dealer’s credit spread increases?

Expected Exposure on Pair of Offsetting Interest Rate Swaps and a Pair of Offsetting Currency Swaps with No Collateral Posted (Figure 18.1)



Interest Rate vs Currency Swaps

- The q_i's are the same for both
- The v_i's for an interest rate swap are on average much less than the v_i's for a currency swap
- The expected cost of defaults on a currency swap is therefore greater.

Simple Example: Single transaction which provides a payoff at time T years and always has positive value to dealer

- CVA has the effect of multiplying value of transaction by e^{-sT} where s is spread between T-year bond issued by counterparty and risk-free T-year bond
- DVA is zero

Futures option

- When exercising the call, we get a long futures position + cash (F - K)
- When exercising the put, we get a short futures position + cash (K - F)

Pricing:

- Black model
- Underlying asset is the futures position (traded product).
- Price process of the underlying asset is GBM.
- Dynamic replication, risk free portfolio.

Black model

Price evolution:

$$dS = \mu S dt + \sigma S dw$$
$$dF = (\mu - r) F dt + \sigma F dw$$

- Futures position is set into the replicating portfolio.
- In the partial differential equation, the delta term does not appear because the value of the futures position is zero.
- Solution for the call:

$$call = e^{-rt}[FN(d_1) - KN(d_2)]$$

$$d_1 = \frac{\ln\left(\frac{F}{K}\right) + 0{,}5\sigma^2 t}{\sigma\sqrt{t}} \qquad d_2 = d_1 - \sigma\sqrt{t}$$

Application of Black's model

Originally for pricing of options for commodity futures positions (Black 1976).

It can be applied for any products if:

- the price of the underlying asset (V) is lognormally distributed in T
- The standard deviation of $\ln(V_T)$ equals to $\sigma\sqrt{t}$
- Interest rates are non-stochastic

Dealer Has Single Uncollateralized Long Forward with Counterparty (Section 18.7.3)

- The value of a long forward contract at time t is $(F_t - K)e^{-r(T-t)}$
- The exposure at time t is $e^{-r(T-t)}\max(F_t - K, 0)$
- Using the Black-Scholes-Merton result to calculate the present value of $E[\max(F_t - K, 0)]$ gives

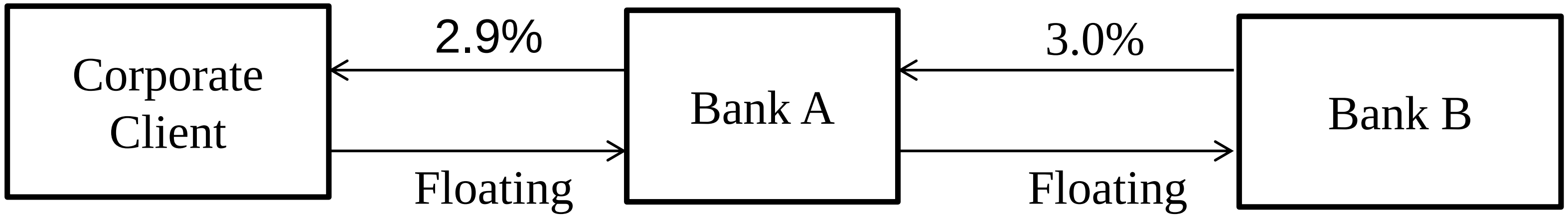
$$v_i = e^{-rT}\left[F_0 N(d_{1,i}) - KN(d_{2,i})\right]$$

where

$$d_{1,i} = \frac{\ln(F_0 / K) + \sigma^2 t_i / 2}{\sigma\sqrt{t_i}} \qquad d_{2,i} = d_{1,i} - \sigma\sqrt{t_i}$$

FVA

FVA, Funding Value Adjustment, considers the situation where a bank enters into a transaction with a client where no margin is posted and hedges this by entering into an offsetting transaction with another bank, e.g.:



FVA continued

- Swap with bank will be cleared through a CCP and bank will post initial margin plus variation margin
- Swap with client requires no margin

FVA continued

- If the swap with Bank B is out-of-the-money and the swap with the client is in the money, the bank is paying variation margin to the CCP but not receiving variation margin from the client
- Cost of funding the variation margin (net of the interest received on the margin) gives rise to a positive FVA

FVA continued

- If we are in the opposite position where the swap with Bank B is in-the-money and the swap with the client is out-of-the-money, the bank is receiving variation margin from the CCP but not paying variation margin to the client
- In this case the variation margin is a source of funding and FVA is negative
- Note: FVA can be calculated on a transaction-by-transaction basis by using Monte Carlo simulation to explore alternative outcomes

MVA

- MVA, margin value adjustment, is the cost of the initial margin.
- This is the difference between the bank's funding cost and the interest paid on the initial margin

Theoretical Arguments

- Many banks use their average debt funding cost to calculate the cost of FVA and MVA
- Financial economic theory would suggest that the initial margin and variation margin are low risk investments. The required return is less than the bank's average funding cost
- Consider the following situation:
  - A bank's average debt funding cost is fed funds rate plus 100 basis points.
  - Because margin is a low risk investment, the return required on margin is the fed funds rate plus 10 bps.
  - The interest earned on margin is the fed funds rate minus 20 basis points

KVA

- KVA is the capital valuation adjustment.
- This is an adjustment for the incremental capital requirements associated with a derivative
- Many banks consider the cost of incremental capital to be the required return on equity
- Financial economists would argue that additional equity makes the bank less risky and should reduce the required return on both debt and equity

KVA

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Calculation Issues

- The calculation of FVA, MVA, and KVA, like the calculations of CVA and DVA, involve very time-consuming Monte Carlo simulations.
- This is because it is necessary to calculate
  - expected future variation and initial margin requirements (in the case of FVA and MVA)
  - expected future capital requirements (in the case of KVA)

Literature

- John C. Hull: Risk Management and Financial Institutions, 6th edition, Chapters 6, 18, 19.
- Basel Committee on Banking Supervision: The Basel Framework
- Black, Fischer. "The pricing of commodity contracts." *Journal of Financial Economics* 3, no. 1-2 (1976): 167-179.